K Shiv Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	7.60
Bhilai Nair Samajam, Sec-8, Bhilai (C.G.)	Class XII (CGBSE)	2011	63.60%
Bhilai Nair Samajam, Sec-8, Bhilai (C.G.)	Class X (CBSE)	2009	80.00%

SUMMARY

Software/Systems Engineer with a strong foundation in networking and low-level systems, combining 1.8 years of experience in industry with more than 4 years of research in programmable data planes (P4 and eBPF) and compiler framework (LLVM). Experienced in C/C++, Python, Linux Networking Stack, Host Networking, Linux Operating System, and Systems programming, with a proven ability to build solutions for real-world systems challenges.

SKILLS

Programming & Compiler Tech: C/C++/Python, eBPF (TC/XDP), P4 Language, LLVM/Clang, Systems Programming

Networking & Protocols: Host/Core Networking, TCP/IP Stack, SDN Concepts, Linux Networking

Systems & Infrastructure: Linux Kernel, Docker/Kubernetes, Vagrant/VirtualBox, DevOps Fundamentals

Tools & Collaboration: GitHub, SQL, Technical Documentation

PROJECTS

"Yaksha-Prashna: Understanding eBPF Bytecode Network Function Behaviour" | CoNEXT 2026 | arXiv '2026: (Link)

- Developed a system to validate the correctness and interactions of third-party eBPF network function bytecode.
- Applied program analysis to extract functional properties of eBPF network functions from the bytecode.
- **Tools & Frameworks:** C++, STL library, LLVM, eBPF/XDP, Libbpf, Linux kernel, bpftool, bpfman.

"In-Network Probabilistic Monitoring Primitives under the Influence of Adversarial Network Inputs" | APNET '23: (Link)

- Analyzed vulnerabilities of monitoring primitives in programmable data planes against adversarial inputs.
- Demonstrated FlowRadar pollution attacks where a few malicious flows corrupted telemetry data.
- Quantified attack impact, revealing a 99% drop in FlowRadar's accuracy under targeted adversarial conditions.
- **Tools & Frameworks:** Python, Wireshark, MS Excel.

"DBVal: Validating P4 Data Plane Runtime Behaviour" | SOSR '21: (Link)

- Developed DBVal, a runtime validation system to detect packet-processing errors in P4 data planes.
- Implemented assertion-based tracking of tables and actions, enabling line-rate validation of runtime Behaviour.
- **Tools & Frameworks:** Python, P4 language, Lex & Yacc, BMv2, Mininet, Wireshark, VirtualBox.

PROFESSIONAL EXPERIENCE

Application Development Analyst (Accenture, Pune, India)

- Integrated a Learning Management System and developed an off-boarding feature for the client's Salesforce environment, enhancing user onboarding and lifecycle workflows.
- **Award: Apex Award -** Extra Miller, in Accenture India.

CERTIFICATIONS

- **Intel Connectivity Academy Level 1** | Issued by: Intel (Link)

RELEVANT COURSES

Advanced Computer Networks, Network Security, Topics in Networks, Research Methods, Introduction to Compiler Engineering, Introduction to Program Analysis and Optimization.

VOLUNTEERING EXPERIENCE

M.S, IIT Hyderabad

- Teaching Assistant for courses Computer Networks and Topics in Networks; authored lab assignments on programmable data planes (P4 and eBPF) and contributed to exam preparation.
- Volunteered in 'International Conference on Distributed Computing and Networking'; managed registrations and session logistics for 200+ participants (IIT Hyderabad, 2025). (Link)
- Volunteered in 'ACM Workshop on Research Opportunities in Computer Science'; coordinated participant registration for 150+ conference delegates (IIT Hyderabad, 2024). (Link)

M.Tech, IIIT Naya Raipur

- Teaching Assistant for Software Defined Networks course; assisted in exam preparation and invigilation.

AWARDS AND HONORS

- ACM SIGMETRICS Student grant: For presenting our ongoing research on "Detecting Adversarial Attacks on Bloom Filters in P4 Data Plane Systems" (2022). (Link)
- COMSNETS Travel Grant: For attending the Conference (2022). (Link)